
Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

Mathematics I

BCA — Semester 1 — 2019 — Answer Sheet

Subject Code: CACS104

Group A

Attempt ALL the questions. $[10 \times 1 = 10]$

1. i) If $A = [-1, 3)$ and $B = [2, 5]$, then $A - B$ is equal to
a) $[-1, 2)$ b) $[-1, 3)$ c) $(-1, 2)$ d) $[-1, 3]$
- ii) If $f(x) = x$ and $g(x) = x + 1$, then what is the value of $(f \circ g)(x)$?
a) $1 + x$ b) $x + 1$ c) $1/(4 + x)$ d) $2 + x$
- iii) What is the reciprocal of the complex number $(2, 1)$?
a) $(1/5, 1/5)$ b) $(2/5, -1/5)$ c) $(-2/5, 1/5)$ d) $(-2, -1)$
- iv) What type of function $f(x) = ax^2 + bx + c$ is?
a) Constant b) Linear c) Identity d) Quadratic
- v) Geometrical meaning of scalar triple product of three vectors a, b, c is the
a) Volume of parallelepiped formed by a, b, c as adjacent sides
b) $\|a\| \times$ Projection of b on a and c
c) $\|b\| \times$ Projection of b on a
d) $\|a \times b \times c\|$
- vi) If a, b, c is in H.P then, what is the value of b ?
a) $2ca/(c+a)$ b) ac c) $ca/(a+c)^2$ d) $ca/(a+c)$
- vii) Which of the following is the rank of the Matrix $\begin{bmatrix} 4 & 2 \\ 4 & 2 \end{bmatrix}$?
a) 0 b) 1 c) 2 d) 3
- viii) In how many ways 6 persons can seat in a round table?
a) 720 b) 360 c) 120 d) 60
- ix) Let $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, and a map $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(x) = A(x)$, then what is the image of $u = [2, 1]$ under T ?
a) $[2, 1]$ b) $[1, 1]$ c) $[2, 0]$ d) $[1, 2]$
- x) If $r = 1/(1 + \cos \theta)$, then this is the equation of..
a) Parabola b) Hyperbola c) Ellipse d) Circle

[10 × 1 = 10]

Answer: Answers:

i) **a) $[-1, 2)$** — $A - B$ = elements in A but not in B . $A = [-1, 3)$, $B = [2, 5]$. So $A - B = [-1, 2)$.

ii) **c) $1/(4 + x)$** — Wait, $(f \circ g)(x) = f(g(x)) = f(x + 1) = x + 1$. But if $f(x) = 1/x$, then $f(g(x)) = 1/(x + 1)$. Given the options, the correct answer based on standard interpretation is **b) $x + 1$** .

iii) **b) $(2/5, -1/5)$** — Reciprocal of $z = 2 + i$ is $1/(2 + i) = (2 - i)/((2 + i)(2 - i)) = (2 - i)/5 = (2/5, -1/5)$.

iv) **d) Quadratic function** — $f(x) = ax^2 + bx + c$ is the standard form of a quadratic function, where the highest power of x is 2.

v) **a) Volume of parallelepiped formed by a, b, c as adjacent sides** — The scalar triple product $[a \ b \ c] = a \cdot (b \times c)$ gives the volume of the parallelepiped with a, b, c as coterminal edges.

vi) **a) $2ca/(c+a)$** — For three numbers in H.P (Harmonic Progression), the middle term $b = 2ac/(a+c)$. This is the harmonic mean of a and c .

vii) b) 1 — Matrix $\begin{bmatrix} 4 & 2 \\ 4 & 2 \end{bmatrix}$ has proportional rows (Row 2 = Row 1). Determinant = $4 \times 2 - 2 \times 4 = 0$. Since it's a non-zero matrix, rank = 1.

viii) c) 120 — Circular permutation of n objects = $(n-1)!$. For 6 persons: $(6-1)! = 5! = 120$.

ix) a) [2,1] — $T(u) = A \cdot u = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. Since A is the identity matrix, T preserves the vector.

x) a) Parabola — $r = 1/(1+\cos \theta)$ is the polar equation of a parabola with eccentricity $e=1$ and focus at the pole.

Group B

Attempt any SIX questions. $[6 \times 5 = 30]$

2. In a class of 100 students, 40 students failed in Mathematics, 70 failed in English and 20 failed in both subjects. Find

- How many students passed in both subjects?
- How many students passed in Mathematics only?
- How many students failed in mathematics only?

[5]

Answer: Solution: Using set theory and Venn diagrams.

Given:

- Total students $n(U) = 100$
- Failed in Math $n(M) = 40$
- Failed in English $n(E) = 70$
- Failed in both $n(M \cap E) = 20$

Step 1: Find students who failed in Math only:

$$n(M \text{ only}) = n(M) - n(M \cap E) = 40 - 20 = 20$$

Step 2: Find students who failed in English only:

$$n(E \text{ only}) = n(E) - n(M \cap E) = 70 - 20 = 50$$

Step 3: Find total students who failed in at least one subject:

$$n(M \cup E) = n(M) + n(E) - n(M \cap E) = 40 + 70 - 20 = 90$$

Step 4: Find students who passed both subjects:

$$\text{Passed both} = n(U) - n(M \cup E) = 100 - 90 = 10 \text{ students}$$

Answers:

- 10 students** passed in both subjects.
- Passed in Math only = Students who failed English only = Total - $n(E) = 100 - 70 = 30$ students.
(Alternatively: Math pass = $100 - 40 = 60$; Math only pass = $60 - 10 = 50$... wait)
Correct: Students who passed Math = $100 - 40 = 60$. Passed Math only = $60 - 10 = 50$ students.
- 20 students** failed in Mathematics only.

3. Find the domain and range of the function $f(x) = (2x+1)/(3-x)$

[5]

Answer: Solution:

Given: $f(x) = (2x+1)/(3-x)$

Domain:

The function is defined for all real numbers except where the denominator is zero.

$$3 - x = 0 \rightarrow x = 3$$

Therefore, **Domain** = $\mathbb{R} - \{3\}$ or $(-\infty, 3) \cup (3, \infty)$

Range:

$$\text{Let } y = (2x+1)/(3-x)$$

$$y(3-x) = 2x+1$$

$$3y - xy = 2x + 1$$

$$3y - 1 = 2x + xy$$

$$3y - 1 = x(2 + y)$$

$$x = (3y - 1)/(y + 2)$$

For x to be defined: $y + 2 \neq 0 \rightarrow y \neq -2$

Therefore, **Range** = $\mathbb{R} - \{-2\}$ or $(-\infty, -2) \cup (-2, \infty)$

Alternative method for Range:

$$\text{As } x \rightarrow 3^-, f(x) \rightarrow +\infty$$

$$\text{As } x \rightarrow 3^+, f(x) \rightarrow -\infty$$

$$\text{As } x \rightarrow \pm\infty, f(x) \rightarrow -2$$

The horizontal asymptote is $y = -2$, which is never reached.

Hence **Range** = $\mathbb{R} - \{-2\}$.

4. Find the Maclaurin series of the function $f(x) = \sin x$.

[5]

Answer: Solution:

Maclaurin series: $f(x) = f(0) + xf'(0) + (x^2/2!)f''(0) + (x^3/3!)f'''(0) + \dots$

Given: $f(x) = \sin x$

Derivatives at $x = 0$:

$$f(x) = \sin x \rightarrow f(0) = 0$$

$$f'(x) = \cos x \rightarrow f'(0) = 1$$

$$f''(x) = -\sin x \rightarrow f''(0) = 0$$

$$f'''(x) = -\cos x \rightarrow f'''(0) = -1$$

$$f^{(4)}(x) = \sin x \rightarrow f^{(4)}(0) = 0$$

$$f^{(5)}(x) = \cos x \rightarrow f^{(5)}(0) = 1$$

Pattern: 0, 1, 0, -1, 0, 1, 0, -1, ...

Maclaurin Series:

$$\sin x = 0 + x(1) + (x^2/2!)(0) + (x^3/3!)(-1) + (x^4/4!)(0) + (x^5/5!)(1) + \dots$$

$$\sin x = x - x^3/3! + x^5/5! - x^7/7! + x^9/9! - \dots$$

Or in summation form:

$$\sin x = \sum_{n=0}^{\infty} (-1)^n x^{2n+1}/(2n+1)!$$

First few terms:

$$\sin x = x - x^3/6 + x^5/120 - x^7/5040 + x^9/362880 - \dots$$

5. Prove that $|1 \ x \ x^2 \mid 1 \ y \ y^2 \mid 1 \ z \ z^2| = (x-y)(y-z)(z-x)$

[5]

Answer: Proof:

$$\text{LHS} = \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix}$$

Apply $R_1 \rightarrow R_1 - R_2$ and $R_1 \rightarrow R_1 - R_3$:

$$= \begin{vmatrix} 1-x & x-x & x^2-x^2 \\ 0 & y-x & y^2-x^2 \\ 0 & z-x & z^2-x^2 \end{vmatrix}$$

Expand along C_1 :

$$= 1 \cdot \begin{vmatrix} y-x & y^2-x^2 \\ z-x & z^2-x^2 \end{vmatrix}$$

$$= (y-x)(z^2-x^2) - (z-x)(y^2-x^2)$$

Factor difference of squares:

$$= (y-x)(z-x)(z+x) - (z-x)(y-x)(y+x)$$

$$= (y-x)(z-x)[(z+x) - (y+x)]$$

$$= (y-x)(z-x)(z+x-y-x)$$

$$= (y-x)(z-x)(z-y)$$

$$= -(x-y) \cdot -(x-z) \cdot (z-y)$$

$$= (x-y)(x-z)(z-y)$$

$$= (x-y) \cdot -(z-x) \cdot -(y-z)$$

$$= (x-y)(y-z)(z-x) = \text{RHS}$$

Hence proved.

6. Find a unit vector perpendicular to the plane containing points $P(1, -1, 0)$, $Q(2, 1, -1)$ and $R(-1, 1, 2)$.

[5]

Answer: Solution:

Given points: $P(1, -1, 0)$, $Q(2, 1, -1)$, $R(-1, 1, 2)$

Step 1: Find vectors PQ and PR

$$PQ = Q - P = (2-1, 1-(-1), -1-0) = (1, 2, -1)$$

$$PR = R - P = (-1-1, 1-(-1), 2-0) = (-2, 2, 2)$$

Step 2: Find normal vector using cross product

$$n = PQ \times PR = \begin{vmatrix} i & j & k \\ 1 & 2 & -1 \\ -2 & 2 & 2 \end{vmatrix}$$

$$\begin{vmatrix} 1 & 2 & -1 \\ -2 & 2 & 2 \end{vmatrix}$$

$$\begin{vmatrix} 1 & 2 & -1 \\ -2 & 2 & 2 \end{vmatrix}$$

$$= i(2 \cdot 2 - (-1) \cdot 2) - j(1 \cdot 2 - (-1) \cdot (-2)) + k(1 \cdot 2 - 2 \cdot (-2))$$

$$= i(4 + 2) - j(2 - 2) + k(2 + 4)$$

$$= 6i - 0j + 6k$$

$$n = (6, 0, 6)$$

Step 3: Find unit vector

$$|n| = \sqrt{6^2 + 0^2 + 6^2} = \sqrt{36 + 0 + 36} = \sqrt{72} = 6\sqrt{2}$$

$$\text{Unit vector} = n/|n| = (6, 0, 6)/(6\sqrt{2}) = (\mathbf{1/\sqrt{2}, 0, 1/\sqrt{2}})$$

$$\text{Or rationalized: } (\mathbf{\sqrt{2}/2, 0, \sqrt{2}/2})$$

The unit vector perpendicular to the plane is $(\mathbf{1/\sqrt{2}, 0, 1/\sqrt{2}})$ or its negative $(\mathbf{-1/\sqrt{2}, 0, -1/\sqrt{2}})$.

7. In how many ways can the letters of word "Sunday" be arranged? How many of these arrangements begin with S? How many begin with S and don't end with y?

[5]

Answer: Solution:

Word: SUNDAY

Total letters = 6 (all distinct: S, U, N, D, A, Y)

1. Total arrangements:

Number of ways = $6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$

2. Arrangements beginning with S:

Fix S at first position. Remaining 5 letters can be arranged in $5!$ ways.

Number of ways = $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$

3. Arrangements beginning with S and not ending with Y:

Fix S at first position. We have 5 remaining positions.

Total arrangements with S at first = $5! = 120$

Arrangements with S at first AND Y at last = $4! = 24$ (fix S and Y, arrange middle 4)

Arrangements with S at first and NOT Y at last = $120 - 24 = 96$

Answers:

- Total arrangements = **720**
- Beginning with S = **120**
- Beginning with S and not ending with Y = **96**

8. If $x+iy = \sqrt{[(1+i)/(1-i)]}$, then show that $x^2+y^2 = 1$.

[5]

Answer: Proof:

Given: $x + iy = \sqrt{[(1+i)/(1-i)]}$

Step 1: Simplify $(1+i)/(1-i)$

Multiply numerator and denominator by $(1+i)$:

$$(1+i)/(1-i) \times (1+i)/(1+i) = (1+i)^2 / (1-i^2)$$

$$= (1 + 2i + i^2) / (1 - (-1))$$

$$= (1 + 2i - 1) / 2$$

$$= 2i / 2$$

$$= i$$

Step 2: Find square root of i

$$x + iy = \sqrt{i}$$

Square both sides:

$$(x + iy)^2 = i$$

$$x^2 + 2ixy + i^2y^2 = i$$

$$x^2 - y^2 + 2ixy = 0 + i$$

Equating real and imaginary parts:

$$\text{Real: } x^2 - y^2 = 0 \rightarrow x^2 = y^2$$

$$\text{Imaginary: } 2xy = 1$$

Step 3: Find $x^2 + y^2$

From $x + iy = \sqrt{i}$, taking modulus of both sides:

$$|x + iy| = |\sqrt{i}|$$

$$\sqrt{x^2 + y^2} = \sqrt{|i|}$$

$$\sqrt{x^2 + y^2} = \sqrt{1}$$

$$\sqrt{x^2 + y^2} = 1$$

Squaring both sides:

$$x^2 + y^2 = 1$$

Hence proved.

Group C

Attempt any TWO questions. [$2 \times 10 = 20$]

9. a) Define conic section. Find the coordinates of vertices, eccentricity and foci of the ellipse $9x^2 + 4y^2 - 18x - 16y - 11 = 0$. [1+5]

b) If $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by $T(x, y) = (x + y, x, y)$ be the linear transformation, then find matrix associated with linear map T. [4]

[10]

Answer: a) Conic Section:

A conic section is a curve obtained by the intersection of a plane with a double-napped right circular cone. Depending on the angle of intersection, we get:

- **Circle:** Plane perpendicular to axis
- **Ellipse:** Plane at angle $>$ angle of generator ($0 < e < 1$)
- **Parabola:** Plane parallel to generator ($e = 1$)
- **Hyperbola:** Plane at angle $<$ angle of generator ($e > 1$)

Ellipse: $9x^2 + 4y^2 - 18x - 16y - 11 = 0$

Rearranging:

$$9(x^2 - 2x) + 4(y^2 - 4y) = 11$$

$$9(x^2 - 2x + 1 - 1) + 4(y^2 - 4y + 4 - 4) = 11$$

$$9(x-1)^2 - 9 + 4(y-2)^2 - 16 = 11$$

$$9(x-1)^2 + 4(y-2)^2 = 36$$

$$(x-1)^2/4 + (y-2)^2/9 = 1$$

This is a vertical ellipse (since $9 > 4$).

$$a^2 = 9 \rightarrow a = 3 \text{ (semi-major axis)}$$

$$b^2 = 4 \rightarrow b = 2 \text{ (semi-minor axis)}$$

Center: (1, 2)

Vertices: $(1, 2 \pm 3) = (1, 5)$ and $(1, -1)$

Eccentricity:

$$c^2 = a^2 - b^2 = 9 - 4 = 5 \rightarrow c = \sqrt{5}$$

$$e = c/a = \sqrt{5}/3 \approx \mathbf{0.745}$$

Foci: $(1, 2 \pm \sqrt{5}) = (1, 2 + \sqrt{5})$ and $(1, 2 - \sqrt{5})$

b) Matrix of Linear Transformation T:

$T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by $T(x, y) = (x + y, x, y)$

Standard basis of $\mathbb{R}^2$: $e_1 = (1, 0)$, $e_2 = (0, 1)$

$$T(e_1) = T(1, 0) = (1 + 0, 1, 0) = (1, 1, 0)$$

$$T(e_2) = T(0, 1) = (0 + 1, 0, 1) = (1, 0, 1)$$

Matrix $A = [T(e_1) \mid T(e_2)]$

$A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$ (3×2 matrix)

10. Define irrational number. Prove that $\sqrt{2}$ is an irrational number. [1+4]

If functions $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 2x + 1$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ defined by $g(x) = x^2 - 2$. Find the formulae for composite functions $f \circ g$ and $g \circ f$ and also verify that $f \circ g \neq g \circ f$. [4+1]

[10]

Answer: Irrational Number:

An irrational number is a real number that cannot be expressed as a simple fraction p/q where p and q are integers and $q \neq 0$. Its decimal representation is non-terminating and non-repeating. Examples: $\sqrt{2}$, $\sqrt{3}$, π , e .

Proof that $\sqrt{2}$ is irrational:

Assume, for contradiction, that $\sqrt{2}$ is rational.

Then $\sqrt{2} = p/q$ where p, q are coprime integers (no common factors), $q \neq 0$.

Squaring both sides:

$$2 = p^2/q^2$$

$$p^2 = 2q^2$$

This means p^2 is even, so p must be even (since $\text{odd}^2 = \text{odd}$).

Let $p = 2k$ for some integer k .

Substituting:

$$(2k)^2 = 2q^2$$

$$4k^2 = 2q^2$$

$$q^2 = 2k^2$$

This means q^2 is even, so q must be even.

But if both p and q are even, they share a common factor of 2, contradicting our assumption that p and q are coprime.

Therefore, our assumption is false. $\sqrt{2}$ is irrational.

Composite Functions:

Given: $f(x) = 2x + 1$, $g(x) = x^2 - 2$

$$(f \circ g)(x) = f(g(x))$$

$$= f(x^2 - 2)$$

$$= 2(x^2 - 2) + 1$$

$$= 2x^2 - 4 + 1$$

$$= 2x^2 - 3$$

$$(g \circ f)(x) = g(f(x))$$

$$= g(2x + 1)$$

$$= (2x + 1)^2 - 2$$

$$= 4x^2 + 4x + 1 - 2$$

$$= 4x^2 + 4x - 1$$

Verification that $f \circ g \neq g \circ f$:

Take $x = 1$:

$$(f \circ g)(1) = 2(1)^2 - 3 = -1$$

$$(g \circ f)(1) = 4(1)^2 + 4(1) - 1 = 7$$

Since $-1 \neq 7$, $f \circ g \neq g \circ f$.

In general, $2x^2 - 3 \neq 4x^2 + 4x - 1$ for most values of x , confirming that composition of functions is not commutative.

11. a) If arithmetic mean, geometric mean and harmonic mean between two unequal positive numbers are A, G, H respectively. Then prove that $A > G > H$. [4]

b) What is the relation between permutation and combination of n objects taken r at a time? A committee of 5 is to be constituted from 6 boys and 5 girls. In how many ways can this be done so as to include at least a boy and a girl? [1+5]

[10]

Answer: a) Proof that $A > G > H$:

Let the two positive numbers be a and b ($a \neq b$).

Arithmetic Mean: $A = (a+b)/2$

Geometric Mean: $G = \sqrt{ab}$

Harmonic Mean: $H = 2ab/(a+b)$

Step 1: Prove $A > G$

Consider $(\sqrt{a} - \sqrt{b})^2 > 0$ (since $a \neq b$)

$$a + b - 2\sqrt{ab} > 0$$

$$a + b > 2\sqrt{ab}$$

$$(a+b)/2 > \sqrt{ab}$$

$$A > G \dots (1)$$

Step 2: Prove $G > H$

$$\text{We know } A \cdot H = [(a+b)/2] \cdot [2ab/(a+b)] = ab = G^2$$

$$\text{So } G^2 = A \cdot H$$

Since $A > G$ (from 1), we have:

$$G^2 = A \cdot H > G \cdot H$$

$$G > H \dots (2)$$

From (1) and (2): **$A > G > H$**

Equality holds if and only if $a = b$.

b) Relation between Permutation and Combination:

The relation is: ${}^nP_r = {}^nC_r \times r!$ or ${}^nC_r = {}^nP_r / r!$

Permutation considers order (arrangement), while combination does not (selection).

Committee Problem:

Total people = 6 boys + 5 girls = 11

Committee size = 5

Condition: At least 1 boy AND at least 1 girl

Possible distributions:

- 1 boy + 4 girls
- 2 boys + 3 girls
- 3 boys + 2 girls
- 4 boys + 1 girl

$$\text{Number of ways} = {}^6C_1 \times {}^5C_1 + {}^{15}C_1 \times {}^{10}C_1 + {}^{20}C_1 \times {}^{10}C_1 + {}^{15}C_1 \times {}^5C_1$$

$$= 6 \times 5 + 15 \times 10 + 20 \times 10 + 15 \times 5$$

$$= 30 + 150 + 200 + 75$$

$$= \mathbf{455 \text{ ways}}$$

Alternative Method:

$$\text{Total ways without restriction} = {}^{11}C_2 = 462$$

Subtract invalid cases:

- All boys: ${}^6C_2 = 6$

- All girls: ${}^1C_2 = 1$

$$\text{Valid ways} = 462 - 6 - 1 = \mathbf{455 \text{ ways}}$$

Note: These answers are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.